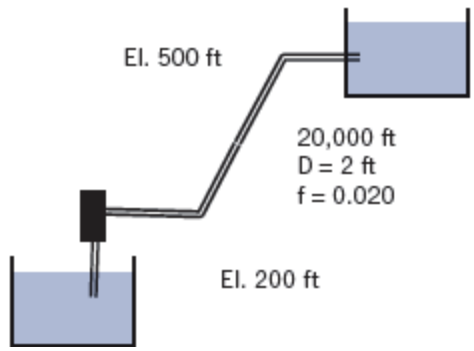
Problem 10.11

(Difficulty 2)

10.11 Using the performance curves in Appendix C, select the smallest diameter Peerless 8AE20G pump operating at 1770 rpm that will deliver a flow of at least 2000 gpm for the pipeline shown. Determine the actual flow rate and the pump electrical power requirement.



Find The smallest pump for this application.

Assumption: The flow is steady and incompressible.

Solution: Use the energy equation to determine the required head and use the pump characteristics to select a pump from Appendix C.

Applying the energy equation to the system between location 1 (the lower reservoir) and 2 (the upper reservoir) we have:

$$z_1 + \frac{p_1}{\gamma} + \frac{V_1^2}{2g} + H_p = z_2 + \frac{p_2}{\gamma} + \frac{V_2^2}{2g} + H_l$$

Where the head loss is given by

$$h_l = f\frac{L}{D}\frac{V^2}{2g}$$

For a desired flow of 2000 gpm, the velocity in the pipe is

$$V = \frac{Q}{A} = \frac{2000\ gpm \times 0.0223\frac{ft^3}{s-gpm}}{\frac{\pi}{4}\times 2} = 1.42\frac{ft}{s}$$

The head loss is

$$h_l = 0.02 \times \frac{20{,}000ft}{2ft} \times \frac{\left(1.42\frac{ft}{s}\right)^2}{2 \times 32.2\frac{ft}{s^2}} = 6.25ft$$

For locations 1 and 2 we have

$$p_1 = p_{2,}\ V_1 = V_2 = 0,\ z_1 = 100\ ft,\ z_2 = 500\ ft$$

The energy equation becomes

$$100\ ft + H_p = 400\ ft + 6.25ft$$

The desired head is then

$$H_p = 400\ ft - 100ft + 6.25ft = 306ft$$

From Appendix C Fig. C.7 for the Peerless 8AE20G pump operating at 1770 rpm, the 18 in. diameter pump will produce a head of 320 ft at a flow rate of 2000 gpm. This is the appropriate pump to select.

The actual flow rate will be slightly larger than 2000 gpm because the head produced is larger. Selecting a large flow rate, determining the new head loss and required head, and then comparing against the pump performance curve will yield the flow rate that matches the head requirement. Through iteration, the actual flow rate will be about 2500 gpm and the total head requirement is 310 ft.

The electrical pump power is given by

$$\dot{W} = \frac{\rho Q H_p}{\eta}$$

The efficiency at 2500 gpm is 79 %. The power is then

$$\dot{W} = \frac{1.94\frac{slug}{ft^3} \times 2500gpm \times 0.0223\frac{ft^3}{s-gpm} \times 310ft \times 32.2\frac{ft}{s^2}}{0.79} = 138{,}000\ \frac{ft-lbf}{s}$$

Or

$$\dot{W} = 251\ hp$$

Problem 10.13 [Difficulty: 2]

10.13 Data from tests of a pump operated at 1450 rpm, with a 12.3-in. diameter impeller, are

Flow rate, Q (cfm)	20	40	60	80	100	120	140
Net positive suction head required, $NPSR$ (ft)	7.1	8.0	8.9	10.3	11.8	12.3	16.9

Develop and plot a curve-fit equation for $NPSHR$ versus volume flow rate in the form $NPSHR = a + bQ^2$, where a and b are constants. If the $NPSHA = 20$ ft, estimate the maximum allowable flow rate of this pump.

Given: Data on a NPSHR for a pump

Find: Curve fit; Maximum allowable flow rate

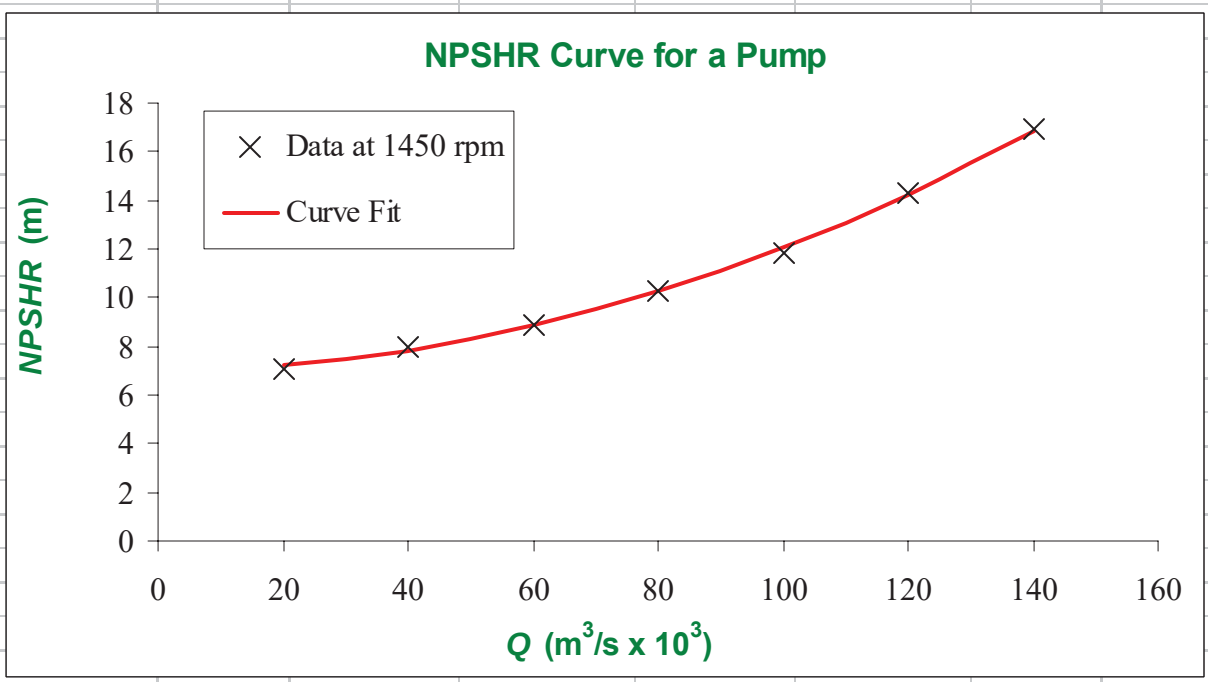
Solution: The results were generated in *Excel*:

Q (cfm)	Q^2	$NPSHR$ (ft)	$NPSHR$ (fit)
20	4.00E+02	7.1	7.2
40	1.60E+03	8.0	7.8
60	3.60E+03	8.9	8.8
80	6.40E+03	10.3	10.2
100	1.00E+04	11.8	12.0
120	1.44E+04	14.3	14.2
140	1.96E+04	16.9	16.9

The fit to data is obtained from a least squares fit to $NPSHR = a + bQ^2$

a =	7.04	ft	Q (cfm)	$NPSHR$ (ft)
b =	5.01E-04	ft/(cfm)^2	160.9	20.00

Use *Goal Seek* to find Q!



Problem 10.16

(Difficulty: 2)

10.16 Part of the water supply for the South Rim of Grand Canyon National Park is taken from the Colorado River [54]. A flow rate of 600 gpm, taken from the river at elevation 3734 ft, is pumped to a storage tank atop the South Rim at 7022 ft elevation. Part of the pipeline is above ground and part is in a hole directionally drilled at angles up to 70° from the vertical; the total pipe length is approximately 13200 ft. Under steady flow operating conditions, the frictional head loss is 290 ft of water in addition to the static lift. Estimate the diameter of the commercial steel pipe in the system. Compute the pumping power requirement if the pump efficiency is 61 percent.

Assumptions: (1) $p_1 = p_2 = p_{atm}$, $\bar{V}_1 = V_2 = 0$ (2) neglect minor losses

Solution: Apply the energy equation with head loss to the total system for steady, incompressible flow using (1) and (2) at inlet and reservoir surface respectively.

$$\frac{p_1}{\rho g} + \alpha_1 \frac{V_1^2}{2g} + z_1 + H_a = \frac{p_2}{\rho g} + \alpha_2 \frac{V_2^2}{2g} + z_2 + \frac{h_{lT}}{g}$$

$$h_{lT} = f \frac{L}{D} \frac{V^2}{2}$$

The pressure and velocities are equal at both the entrance and exit, and the head loss is then given by:

$$H_a = z_2 - z_1 + \frac{h_{lT}}{g}$$

$$\frac{h_{lT}}{g} = f \frac{L}{D} \frac{V^2}{2g} = 290\ ft$$

Since

$$f = f(Re, e/D)$$

and D is unknown, we must iterate. For commercial steel, $e = 0.00015\ ft$,

The iteration procedure is to assume D, calculate V, Re; determine f (Eq.8.37 a,b); calculate h_{lT}/g and compare to value of 290 ft. A table of the guess values for diameter and the resulting head are given below:

D (in.)	V (ft/s)	Re	f_0	$f^{-0.5}$	f	h_{lT}/g (ft)
12	1.70	1.46E+06	0.0139	8.517	0.0138	8.2
10	2.45	1.75E+06	0.0141	8.433	0.0141	20.8
8	3.83	2.19E+06	0.0146	8.306	0.0145	65.3
6	6.81	2.92E+06	0.0153	8.111	0.0152	289

Thus

$$D = 6.0\ in$$

Alternatively, these relations could be programmed into an equation solver and the solution for D computed directly.

The total pump head is:

$$H_a = 7022 - 3734 + 290 = 3578\ ft$$

The pump power is:

$$P_m = \frac{\dot{W}_h}{\eta_p} = \frac{\rho g Q H}{\eta_p}$$

$$P_m = \frac{1}{0.61} \times 62.4\ \frac{lbf}{ft^3} \times 600\ \frac{gal}{min} \times \frac{ft^3}{7.48\ gal} \times \frac{min}{60\ s} \times 3578\ ft \times \frac{hp \cdot s}{550\ ft \cdot lbf}$$

$$P_m = 890\ hp$$